KONERU LAKSHMAIAH EDUCATION FOUNDATION

(Deemed to be University)

Course name: Computational Foundations for Artificial Intelligence

Project title: Hospital bed allocation system

Team members:

2520030012 → Dharmika

2520030211 → Bhaarathi

2520090232 → Sahithya

Under the guidence of

Dr. Ramakanth Komati

Term -3 semester 2025-2026

Abstarct

The Hospital Bed Allocation System is designed to efficiently manage and allocate hospital beds to patients in real time, improving healthcare service delivery and reducing manual errors. In many hospitals, bed management is often handled manually, leading to delays, miscommunication, and inefficient utilization of resources. This system provides a digital platform that tracks bed availability across different departments such as ICU, general ward, and emergency units.

The proposed system enables hospital staff to quickly check bed status, assign beds to incoming patients, and update availability instantly after discharge. It also maintains patient records, prioritizes critical cases, and ensures optimal use of hospital resources. By automating the allocation process, the system reduces waiting time, enhances transparency, and supports better decision-making.

Overall, the Hospital Bed Allocation System improves operational efficiency, ensures timely patient care, and helps hospitals respond effectively during emergencies or high patient inflow situations.